

# implementing and analyzing a basic rl algorithm

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## 1. objective

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this report documents the implementation and analysis of a 5x5 gridworld mdp solved using three approaches: linear system solving, value iteration, and policy iteration.

## 2. environment

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### 2.1 environment parameters

| parameter     | value                         |
|---------------|-------------------------------|
| grid size     | 5x5 (25 cells total)          |
| goal state    | cell 14 (reward +5.0)         |
| danger states | cells 2, 18, 21 (reward -5.0) |
| blocked cells | cells 6, 7, 11, 12            |
| valid states  | 17                            |
| actions       | up, right, down, left         |

### 2.2 mdp components

**state space:** cells 0-24 (row-major), excluding blocked cells 6,7,11,12.

**action space:** four moves; hitting wall/blocked = stay in place.

**transition model:** probability (1-noise) moves intended direction; probability noise moves random.

**reward function:** goal +5.0 (terminal), danger -5.0 (terminal), others 0.0.

## 3. implementation

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### 3.1 gridworld class

the `gridworld` class implements: `__init__`, `_state_from_action`, `is_terminal`, `get_states`, `get_actions`, `get_reward`, `get_transitions`, `solve_linear_system`.

### 3.2 key details

- state indexing: row-major
- noise=0.0: deterministic; noise=0.2: 0.8 intended, 0.067 each other

## 4. methods

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### 4.1 linear system solver

$v = r + \gamma \times p \times v \rightarrow ax = b$  where  $a = i - \gamma p/4$ ,  $b =$  expected rewards. solved via `np.linalg.solve()`.

### 4.2 value iteration

$v(s) \leftarrow \max_a \sum p(s'|s,a)[r(s') + \gamma v(s')]$ . iterate until  $\delta < \text{tolerance}$ .

### 4.3 policy iteration

alternates: policy evaluation ( $v = r + \gamma pv$ ) and policy improvement ( $\pi(s) = \operatorname{argmax}_a q(s,a)$ ).

## 5. results

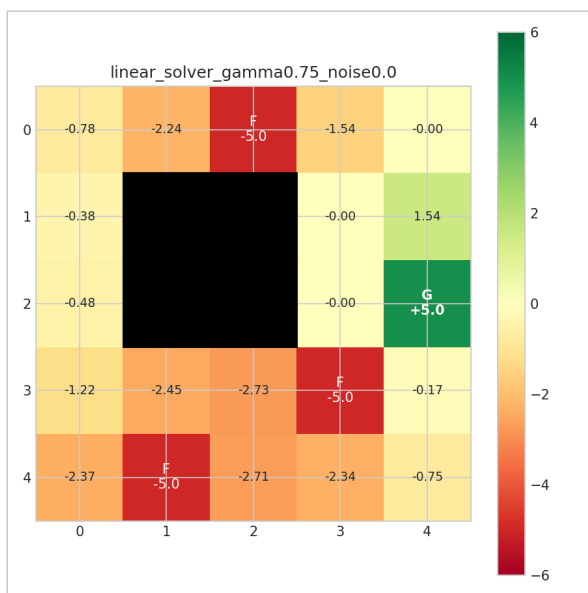
### 5.1 linear system solver

$\gamma = 0.95$ :

```
-2.69  -3.60  -5.00  -1.64  -0.00
-2.35  ----  ----  -0.00  1.64
-2.51  ----  ----  -0.00  goal
-3.19  -3.91  -4.10  -5.00  -0.57
-3.82  -5.00  -3.99  -3.45  -1.82
```

$\gamma = 0.75$ :

```
-0.78  -2.24  -5.00  -1.54  -0.00
-0.38  ----  ----  -0.00  1.54
-0.48  ----  ----  -0.00  goal
-1.22  -2.45  -2.73  -5.00  -0.17
-2.37  -5.00  -2.71  -2.34  -0.75
```



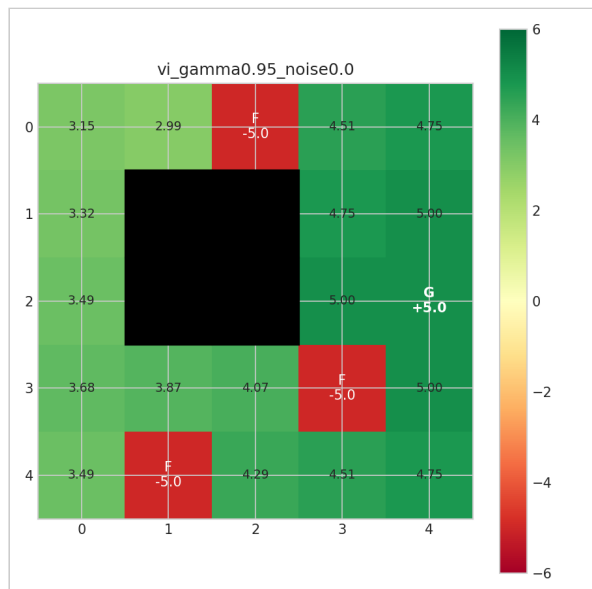
### 5.2 value iteration

| configuration             | iterations |
|---------------------------|------------|
| $\gamma=0.95$ , noise=0.0 | 12         |
| $\gamma=0.75$ , noise=0.0 | 12         |
| $\gamma=0.95$ , noise=0.2 | 15         |
| $\gamma=0.75$ , noise=0.2 | 7          |

$\gamma=0.95$ , noise=0.0:

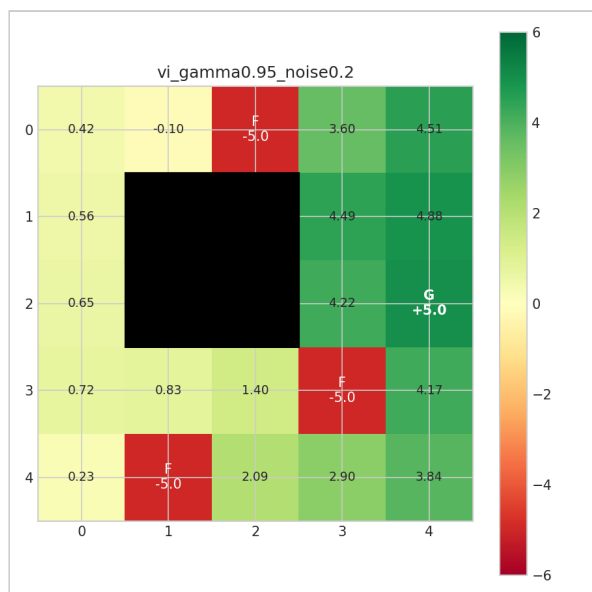
```
3.15  2.99  -5.00  4.51  4.75
3.32  ----  ----  4.75  5.00
3.49  ----  ----  5.00  goal
```

|      |       |      |       |      |
|------|-------|------|-------|------|
| 3.68 | 3.87  | 4.07 | -5.00 | 5.00 |
| 3.49 | -5.00 | 4.29 | 4.51  | 4.75 |



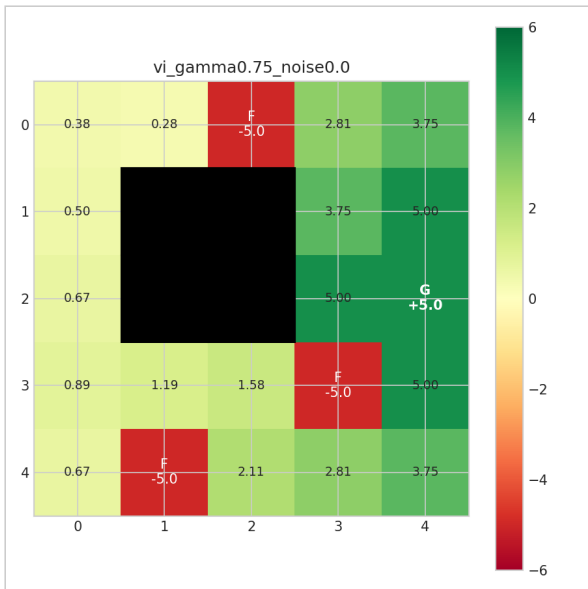
**$\gamma=0.95$ , noise=0.2:**

|      |       |       |       |      |
|------|-------|-------|-------|------|
| 0.42 | -0.10 | -5.00 | 3.60  | 4.51 |
| 0.56 | ----  | ----  | 4.49  | 4.88 |
| 0.65 | ----  | ----  | 4.22  | goal |
| 0.72 | 0.83  | 1.40  | -5.00 | 4.17 |
| 0.23 | -5.00 | 2.09  | 2.90  | 3.84 |



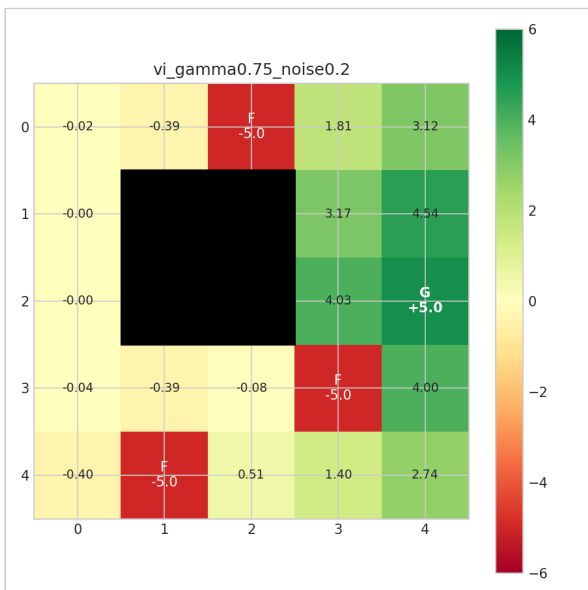
**$\gamma=0.75$ , noise=0.0:**

|      |       |       |       |      |
|------|-------|-------|-------|------|
| 0.38 | 0.28  | -5.00 | 2.81  | 3.75 |
| 0.50 | ----  | ----  | 3.75  | 5.00 |
| 0.67 | ----  | ----  | 5.00  | goal |
| 0.89 | 1.19  | 1.58  | -5.00 | 5.00 |
| 0.67 | -5.00 | 2.11  | 2.81  | 3.75 |



**$\gamma=0.75$ , noise=0.2:**

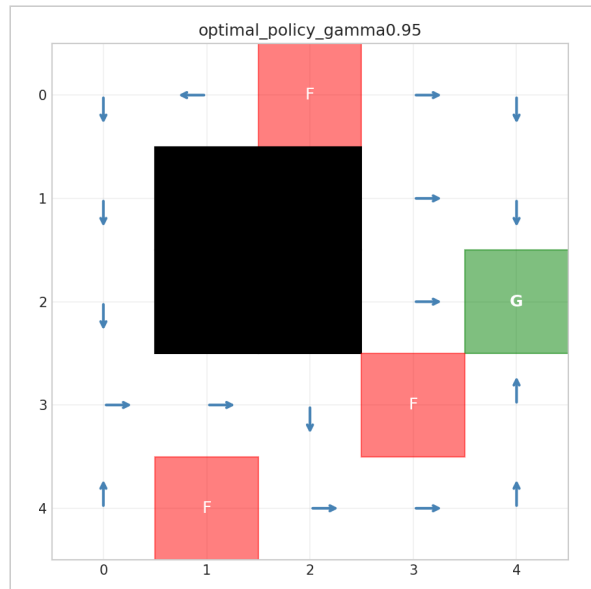
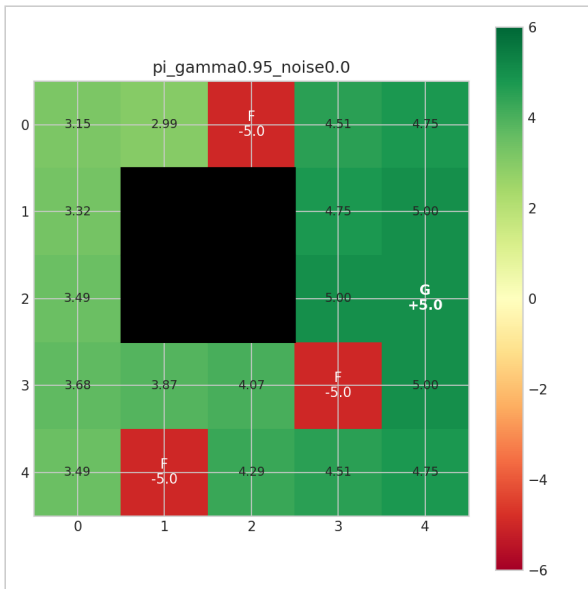
|       |       |       |       |      |
|-------|-------|-------|-------|------|
| -0.02 | -0.39 | -5.00 | 1.81  | 3.12 |
| -0.00 | ----  | ----  | 3.17  | 4.54 |
| -0.00 | ----  | ----  | 4.03  | goal |
| -0.04 | -0.39 | -0.08 | -5.00 | 4.00 |
| -0.40 | -5.00 | 0.51  | 1.40  | 2.74 |



## 5.3 policy iteration

**$\gamma=0.95$ , noise=0.0 (10 iterations):**

|      |       |       |       |      |
|------|-------|-------|-------|------|
| 3.15 | 2.99  | -5.00 | 4.51  | 4.75 |
| 3.32 | ----  | ----  | 4.75  | 5.00 |
| 3.49 | ----  | ----  | 5.00  | goal |
| 3.68 | 3.87  | 4.07  | -5.00 | 5.00 |
| 3.49 | -5.00 | 4.29  | 4.51  | 4.75 |



## 5.4 optimal policy map

| state     | action  | rationale     |
|-----------|---------|---------------|
| 0         | down    | toward goal   |
| 1         | left    | avoid blocked |
| 3,8,13    | right   | approach goal |
| 4,9,19,24 | down/up | reach goal    |

## 6. convergence analysis

| algorithm        | config                    | iterations |
|------------------|---------------------------|------------|
| linear solver    | $\gamma=0.95$             | 1          |
| linear solver    | $\gamma=0.75$             | 1          |
| value iteration  | $\gamma=0.95$ , noise=0.0 | 12         |
| value iteration  | $\gamma=0.75$ , noise=0.0 | 12         |
| value iteration  | $\gamma=0.95$ , noise=0.2 | 15         |
| value iteration  | $\gamma=0.75$ , noise=0.2 | 7          |
| policy iteration | $\gamma=0.95$ , noise=0.0 | 10         |

**factors affecting convergence:** - discount factor  $\gamma$ : higher  $\gamma \rightarrow$  more iterations - noise: increases uncertainty, more iterations - grid topology: blocked cells affect value propagation

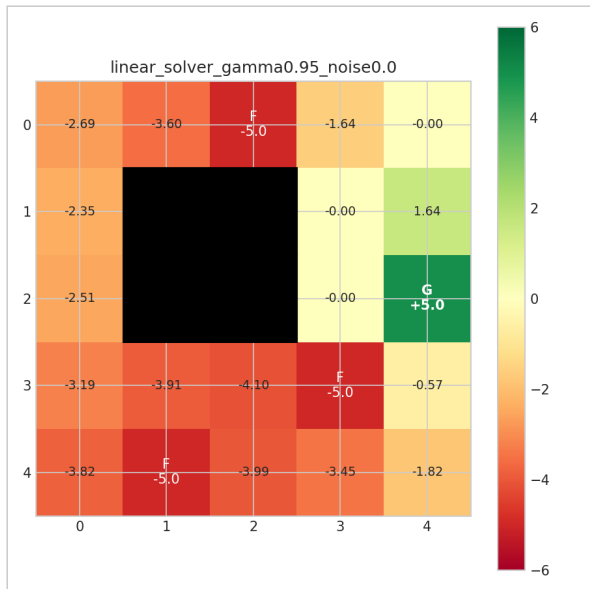
## 7. algorithm comparison

**linear solver:** closed-form, deterministic only, random policy baseline.

**value iteration:** simple, works for stochastic, max over actions each update.

**policy iteration:** fewer outer iterations, explicit policy at each step.

all three produce correct results; vi and pi reach identical optimal policy.



**references:** sutton & barto (2018) reinforcement learning: an introduction - <https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf>